

3.1 Lab Analysis: Newton's Second Law of Motion

A cart on a track was pulled by a hanging mass on a string over a pulley. A motion detector recorded position and velocity. You fit a line to the velocity vs. time graph; its **slope** is the cart's acceleration.

Trial	Cart mass (g)	Hanging mass (g)	Acceleration (m/s ²)
Light, $F = 10$ g	390	10	0.23
Light, $F = 20$ g	390	20	0.47
Light, $F = 50$ g	390	50	1.10
Heavy, $F = 10$ g	890	10	0.098
Heavy, $F = 20$ g	890	20	0.21
Heavy, $F = 50$ g	890	50	0.53

Part 1: How does acceleration depend on force?

Compare trials that use the **same cart** but different hanging masses.

1. Consider the **light cart** (390 g) pulled by 10 g and then by 20 g.

(a) What is the ratio of the hanging masses? $\frac{20}{10} =$

(b) What is the ratio of the accelerations? $\frac{0.47}{0.23} =$

(c) What do you notice about these two ratios?

2. Now compare the **heavy cart** pulled by 10 g and by 50 g.

(a) Ratio of hanging masses: $\frac{50}{10} =$

(b) Ratio of accelerations: $\frac{0.53}{0.098} =$

(c) Is the pattern the same as in question 1?

3. Complete: When the cart mass stays the same, the acceleration is _____ to the force.

Part 2: How does acceleration depend on mass?

Now compare trials that use the **same hanging mass** but different carts.

4. Both carts are pulled by a **20 g** hanging mass.

(a) Ratio of cart masses (heavy to light): $\frac{890}{390} =$

(b) Ratio of accelerations (light to heavy): $\frac{0.47}{0.21} =$

- (c) What do you notice about these two ratios?

5. Verify the pattern with the **50 g** hanging mass.

(a) Ratio of accelerations (light to heavy): $\frac{1.10}{0.53} =$

- (b) Does this confirm the same relationship?

6. Complete: When the force stays the same, the acceleration is *inversely* _____ to the mass.

Part 3: Converting to Newtons — discovering $F = ma$

The hanging mass creates a gravitational force: $F = mg$, where $g = 9.8 \text{ m/s}^2$.

7. Convert each hanging mass to a force in Newtons. (*Convert grams to kilograms first.*)

(a) 10 g: $F =$

(b) 20 g: $F =$

(c) 50 g: $F =$

8. For each trial, divide the force (in Newtons) by the measured acceleration.

(a) Light cart, 20 g: $\frac{F}{a} = \underline{\hspace{2cm}} =$ units: $\underline{\hspace{2cm}}$

(b) Heavy cart, 20 g: $\frac{F}{a} = \underline{\hspace{2cm}} =$ units: $\underline{\hspace{2cm}}$

(c) Heavy cart, 50 g: $\frac{F}{a} = \underline{\hspace{2cm}} =$ units: $\underline{\hspace{2cm}}$

9. Compare your F/a values to the cart masses (converted to kilograms). What do you notice?

10. Write the equation that relates force F , mass m , and acceleration a .

Part 4: Acceleration from the position graph

In Graphical Analysis, you also fit the position data to a quadratic: $x = At^2 + Bt + C$.

11. For constant acceleration, position follows $x = \frac{1}{2}at^2 + v_0t + x_0$.
 - (a) Comparing the two equations, what does A equal in terms of acceleration a ?

 - (b) For the Heavy cart with 20 g, the position fit gave $A = 0.107$.
Calculate the acceleration from this value of A .

 - (c) The velocity slope gave $a = 0.21 \text{ m/s}^2$ for the same trial. Do the two methods agree?